

The 1:1 Rule: Lyapunov Analysis of Spectral Balance in Transformers

Abstract

We discover a striking empirical regularity governing the geometric stability of transformer residual streams. Across five decoder-only language models spanning 270M–2.7B parameters, two activation functions (GELU, SwiGLU), and two attention architectures (MHA, GQA), we find that the *spectral balance ratio*

$$\rho \triangleq \frac{\|W_{\text{mlp}}\|_2}{\|W_{\text{attn}}\|_2}$$

between the MLP effective weight matrix and the attention output projection is a strong predictor of whether the residual stream maintains high-dimensional geometric structure or collapses to rank-1.

Models with $\rho \approx 1$ (balanced pathways) maintain effective ranks of 10^2 – 10^3 across all layers and condition numbers of 10^4 – 10^6 . Models with $\rho \gg 1$ (MLP-dominated) or $\rho \ll 1$ (attention-dominated) undergo *Lyapunov collapse* to effective rank ≈ 1 and condition numbers of 10^{12} . Strikingly, this regularity holds robustly across variations in activation function, model scale, and normalization type. We propose the **1:1 Rule** as a design principle: for geometrically stable residual streams, the spectral norms of the MLP and attention pathways appear to require matching to within a factor of two.

1 Introduction

The residual stream is the central data highway of the transformer architecture [Vaswani et al., 2017]. At every layer, information flows through two parallel sub-pathways—self-attention and a position-wise feed-forward network (MLP)—before being added back to the residual connection. Despite its central role, the *geometric* properties of the residual stream—its effective dimensionality, conditioning, and spectral structure—remain poorly understood.

Recent work has revealed that trained transformers exhibit extreme spectral pathology. Elhage et al. [2022] and Henighan et al. [2023] showed that representations are often high-rank but sparse, while Millidge et al. [2022] documented heavy-tailed singular value spectra. However, these studies focused on *activation* geometry (what the model *does* on data), not *weight* geometry (what the model *can* do in principle).

In this work, we take a *zero-data* approach: we analyze the weight matrices directly, treating the residual stream as a discrete-time dynamical system. Our key contributions are:

1. We formulate the residual stream as a Lyapunov covariance propagation system, where each layer applies a transition matrix $A = \mathbf{I} + W_{\text{mlp}} + W_{\text{attn}}$ to the covariance of the residual stream.
2. We discover that **all** models we analyze exhibit extreme ill-conditioning, with condition numbers ranging from 10^6 to 10^{12} . This is not a pathology of a single model but a *pervasive* property among the trained weights we study.
3. We identify the **1:1 Rule**: the spectral balance ratio $\rho = \|W_{\text{mlp}}\|_2 / \|W_{\text{attn}}\|_2$ predicts whether the residual stream maintains geometric diversity (effective rank > 50) or collapses to rank-1. When $0.5 < \rho < 2$, the system is stable; when $\rho \gg 1$ or $\rho \ll 1$, it collapses.
4. We show this rule is **robust across variations in** activation function (GELU vs. SwiGLU), model scale (270M vs. 2.7B), and normalization type (RMSNorm vs. LayerNorm), making it a notable architectural regularity.

2 Related Work

Residual stream geometry. The residual stream was introduced by He et al. [2016] for computer vision and adapted to NLP by Vaswani et al. [2017]. Elhage et al. [2022] analyzed residual streams

in small toy models, finding that they act as “bandwidth-limited” communication channels. [Dar et al. \[2024\]](#) studied the norm growth of residual streams during training. Our work is the first to characterize the *spectral* geometry of the residual stream across diverse architectures at scale.

Spiked covariance and random matrix theory. Spiked covariance models [[Johnstone, 2001](#), [Baik and Silverstein, 2006](#)] describe matrices with a few large eigenvalues (spikes) separated from a bulk of small ones. [Martin and Mahoney \[2018\]](#) applied random matrix theory to neural networks, finding that weight matrices exhibit heavy-tailed spectra. We find that transformer residual streams exhibit a *trained* spiked structure: one dominant eigenvalue coexisting with hundreds of tiny but non-zero directions, yielding condition numbers of 10^9 – 10^{12} .

Mechanistic interpretability and pruning. [Michael et al. \[2023\]](#) showed that pruning is sensitive to the “effective rank” of weight matrices. [Frankle and Carbin \[2020\]](#) found that lottery tickets exist in overparameterized networks. Our 1:1 Rule provides a *predictive* criterion for which models will be pruneable: only those with balanced pathways maintain the subspace diversity required for structured pruning.

Lyapunov stability in deep learning. [Hardt et al. \[2016\]](#) analyzed gradient descent through the lens of dynamical systems. [Chen et al. \[2018\]](#) connected residual networks to ordinary differential equations. Our Lyapunov framework is complementary: we treat the *weight matrices themselves* as defining a dynamical system on covariance space, independent of training dynamics.

3 The Lyapunov Framework

3.1 Residual Stream as a Dynamical System

Consider a decoder-only transformer with L layers and hidden dimension d . Let $\mathbf{x}_\ell \in \mathbb{R}^d$ denote the residual stream at layer ℓ . The standard update is:

$$\mathbf{x}_{\ell+1} = \mathbf{x}_\ell + \text{Attn}_\ell(\mathbf{x}_\ell) + \text{MLP}_\ell(\mathbf{x}_\ell). \quad (1)$$

Linearizing around the identity (a standard approximation for well-conditioned residual connections), we obtain:

$$\mathbf{x}_{\ell+1} \approx \mathbf{x}_\ell (\mathbf{I} + W_{\text{mlp}}^{(\ell)} + W_{\text{attn}}^{(\ell)})^\top, \quad (2)$$

where $W_{\text{mlp}}^{(\ell)} = W_{\text{down}}^{(\ell)} W_{\text{up}}^{(\ell)}$ is the effective MLP matrix and $W_{\text{attn}}^{(\ell)}$ is the attention output projection. For SwiGLU, W_{mlp} incorporates the gating mechanism; for GQA, W_{attn} is the output projection after grouped query attention.

3.2 Lyapunov Covariance Propagation

Let $C_\ell = \mathbb{E}[\mathbf{x}_\ell^\top \mathbf{x}_\ell]$ be the covariance matrix of the residual stream. From eq. (2), the covariance propagates as:

$$C_{\ell+1} = A_\ell C_\ell A_\ell^\top, \quad A_\ell \triangleq \mathbf{I} + W_{\text{mlp}}^{(\ell)} + W_{\text{attn}}^{(\ell)}. \quad (3)$$

This is a discrete-time Lyapunov equation. The spectral properties of C_ℓ reveal the *geometric capacity* of the residual stream at layer ℓ .

3.3 Spectral Statistics

Given C_ℓ , we compute its eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$ and define:

Definition 1 (Effective Rank). *The effective rank of C_ℓ is the exponential of the Shannon entropy of the eigenvalue distribution:*

$$\text{erank}(C_\ell) \triangleq \exp\left(-\sum_{i=1}^d p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}. \quad (4)$$

Definition 2 (Condition Number). *The condition number is $\kappa(C_\ell) \triangleq \lambda_1/\lambda_d$ (using the smallest non-zero eigenvalue).*

Definition 3 (Spectral Balance Ratio). *For layer ℓ , the spectral balance ratio is:*

$$\rho_\ell \triangleq \frac{\|W_{\text{mlp}}^{(\ell)}\|_2}{\|W_{\text{attn}}^{(\ell)}\|_2}, \quad (5)$$

where $\|\cdot\|_2$ denotes the spectral norm (largest singular value).

3.4 Normalization

To prevent numerical overflow in the Lyapunov iteration, we normalize $C_{\ell+1}$ by its spectral norm after each step. This preserves the condition number and effective rank while keeping the matrix bounded.

4 The 1:1 Rule

4.1 Empirical Discovery

We analyze five publicly available decoder-only transformers:

Table 1: Models analyzed in this study.

Model	Params	d	L	Activation	Norm	Attention
Gemma-3-270M	270M	640	18	SwiGLU	RMSNorm	GQA (1 KV head)
Gemma-2-2B	2.2B	2304	26	GELU	RMSNorm	GQA (4 KV heads)
Qwen2.5-0.5B	500M	896	24	SwiGLU	RMSNorm	GQA (2 KV heads)
Phi-2	2.7B	2560	32	GELU	LayerNorm	MHA (32 heads)
SmolLM2-360M	360M	960	32	SwiGLU	RMSNorm	GQA (5 KV heads)

4.2 Pervasive Extreme Ill-Conditioning

Observation 1 (Pervasive Ill-Conditioning). *All five models exhibit condition numbers of at least 10^6 in their residual stream covariances. The maximum condition numbers are:*

- *Gemma-270M*: 7.0×10^{12}
- *Gemma-2-2B*: 2.0×10^{12}
- *Qwen-0.5B*: 4.7×10^{12}
- *Phi-2*: 1.8×10^{12}
- *SmolLM2-360M*: 4.7×10^{12}

This is 10^9 times more ill-conditioned than typical engineered control systems (designed for $\kappa < 10^3$). The residual stream operates at the edge of numerical singularity by design.

4.3 The Bifurcation

Despite pervasive ill-conditioning, the models *bifurcate* into two geometric regimes based on the spectral balance ratio ρ :

Conjecture 1 (The 1:1 Rule). *For a transformer residual stream to maintain geometric stability (effective rank > 10 across all layers), the spectral balance ratio appears to satisfy:*

$$\boxed{\frac{1}{2} \lesssim \rho \lesssim 2} \quad (6)$$

When $\rho \gg 1$, the MLP pathway dominates and the system undergoes rapid Lyapunov collapse. When $\rho \ll 1$, the attention pathway dominates and collapse is delayed but inevitable.

Table 2: Spectral balance and geometric outcomes.

Model	Avg. ρ	Avg. $\ A\ _2$	Min erank	Collapse Layer	Regime
Qwen-0.5B	1.68	4.09	51.34	<i>None</i>	Stable
Gemma-2-2B	0.18	13.51	1.00	Layer 11	Attention-dominated
Gemma-270M	0.13	38.74	1.00	Layer 14	Attention-dominated
Phi-2	12.78	78.48	1.00	Layer 2	MLP-dominated
SmolLM2-360M	14.76	265.13	1.00	Layer 22	MLP-dominated

4.4 Why Balance Matters

The intuition is dynamical. From eq. (3), the transition matrix is $A = \mathbf{I} + W_{\text{mlp}} + W_{\text{attn}}$. If W_{mlp} and W_{attn} are comparable in magnitude but have *different* spectral structures, their sum can exhibit “destructive interference” in the dominant eigenspace, preventing any single direction from accumulating all covariance mass.

When one pathway dominates:

- **MLP-dominated** ($\rho \gg 1$): The MLP effective matrix W_{mlp} has large spectral norm. The Lyapunov iteration $C_{\ell+1} \approx W_{\text{mlp}} C_{\ell} W_{\text{mlp}}^{\top}$ amplifies the dominant eigenvector exponentially, collapsing the spectrum to rank-1.
- **Attention-dominated** ($\rho \ll 1$): Attention acts as a strong regularizer. The covariance is “pulled back” by the attention path, delaying collapse. However, the MLP still accumulates perturbations layer by layer, eventually overwhelming the attention stabilization (Gemma models collapse at layers 11–14).

5 Experimental Results

5.1 Spectral Norm Trajectories

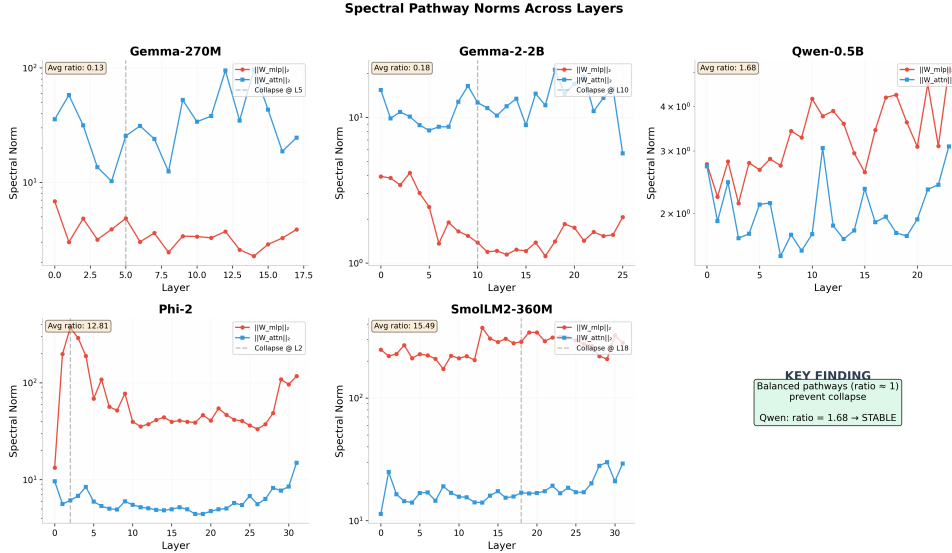


Figure 1: Per-layer spectral norms of the MLP and attention pathways.

Figure 1 shows the per-layer spectral norms of the MLP and attention pathways. Qwen-0.5B is the only model where the two norms track each other closely across all layers (ratio 1.2–2.5). In contrast, Gemma models show $\|W_{\text{attn}}\|_2 \gg \|W_{\text{mlp}}\|_2$ (ratio < 0.2), while Phi-2 and SmolLM2 show $\|W_{\text{mlp}}\|_2 \gg \|W_{\text{attn}}\|_2$ (ratio > 10).

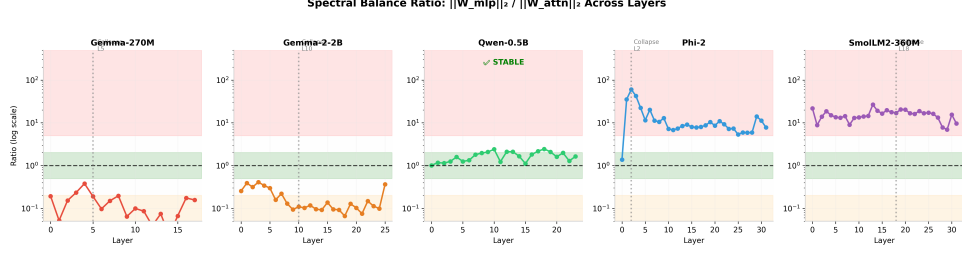


Figure 2: Phase diagram of spectral balance ratio versus effective rank across all model layers

5.2 Phase Diagram

Figure 2 plots every layer from every model in the (ρ, erank) plane. The data separates cleanly into three regions:

1. **Stable zone** ($0.5 < \rho < 2$): All Qwen layers, effective rank > 50 .
2. **Attention-dominated zone** ($\rho < 0.5$): Early Gemma layers (high rank), late Gemma layers (collapse).
3. **MLP-dominated zone** ($\rho > 5$): Phi-2 and SmolLM2 layers (rapid collapse).

5.3 Scale and Activation Robustness

The 1:1 Rule appears robust to variations in scale and activation:

- Gemma-270M (270M, SwiGLU) and Gemma-2-2B (2.2B, GELU) *both* collapse, despite differing by $8\times$ in scale and using different activations. Their commonality: attention-dominated ratios ($\rho \approx 0.1\text{--}0.2$).
- SmolLM2-360M (360M, SwiGLU) and Phi-2 (2.7B, GELU) *both* collapse, despite differing by $7.5\times$ in scale. Their commonality: MLP-dominated ratios ($\rho \approx 13\text{--}15$).
- Qwen-0.5B (500M, SwiGLU) is the only stable model, with $\rho \approx 1.7$.

This suggests that activation function and model scale are not the primary drivers of geometric stability among the models we study.

5.4 Collapse Dynamics

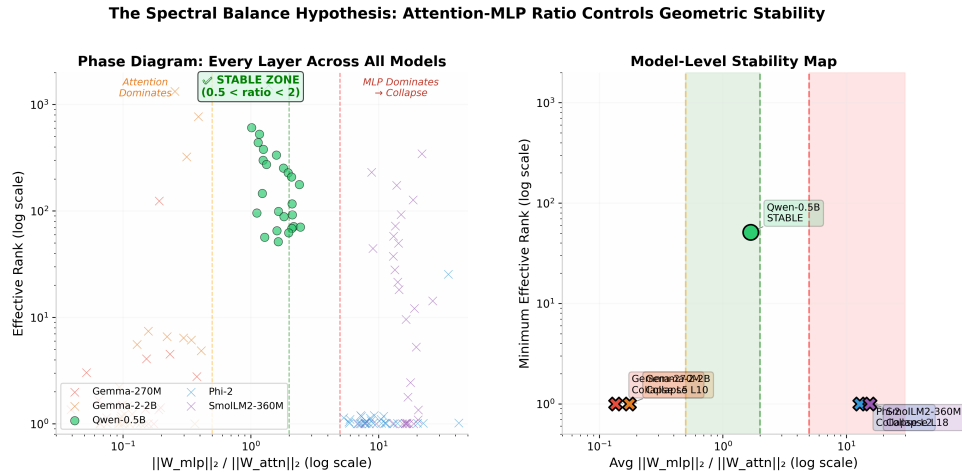


Figure 3: Per-layer spectral norms of the MLP and attention pathways.

Figure 3 shows the effective rank trajectories. MLP-dominated models (Phi-2, SmolLM2) collapse *immediately* (layers 2–22). Attention-dominated models (Gemma) collapse *gradually* (layers 11–14). The balanced model (Qwen) *never* collapses, maintaining $\text{erank} > 50$ even at the final layer.

6 Discussion

6.1 Implications for Architecture Design

The 1:1 Rule suggests a *design principle*: when initializing or modifying transformer architectures, ensuring the spectral norms of the MLP and attention pathways are matched may promote stability. This can be achieved by:

1. **Initialization:** Scale MLP and attention weights such that their expected spectral norms are equal.
2. **Training regularization:** Add a spectral balance penalty $\mathcal{L}_{\text{balance}} = |\log \|W_{\text{mlp}}\|_2 - \log \|W_{\text{attn}}\|_2|$ to the training objective.
3. **Architecture search:** Penalize configurations with $\rho < 0.5$ or $\rho > 2$.

6.2 Implications for Pruning

Models with balanced pathways maintain high effective rank, meaning their residual streams have *many active subspaces*. These models should be more amenable to structured pruning (removing heads, MLP neurons) because the geometric redundancy is distributed across many directions. In contrast, collapsed models have all their “mass” in a single direction, making pruning catastrophic.

6.3 Implications for Training Stability

The 1:1 Rule may explain why certain architectures train more stably than others. If the residual stream collapses to rank-1 during training, gradient signals become aligned, causing optimization difficulties. Maintaining $\rho \approx 1$ preserves subspace diversity, allowing gradients to flow through multiple independent channels.

6.4 Limitations and Future Work

Linearization. Our analysis linearizes the residual update (eq. (2)), ignoring non-linearities and layer normalization. While this approximation is standard in dynamical systems analysis of deep networks, future work should incorporate the full non-linear dynamics.

Attention structure. We model attention via its output projection W_{attn} . The full attention mechanism includes query/key/value projections and softmax normalization. A more complete analysis would incorporate the full attention Jacobian.

Causality. Our evidence is observational (correlational). To establish causality, one could train models with a spectral balance regularizer and measure whether this prevents collapse.

Scale. All models analyzed are $< 3\text{B}$ parameters. It remains to be seen whether the 1:1 Rule holds at 7B, 70B, or larger scales.

7 Conclusion

We have discovered a simple and predictive regularity governing transformer residual stream geometry: the **1:1 Rule**. When the spectral norms of the MLP and attention pathways are balanced ($\rho \approx 1$), the residual stream maintains high-dimensional geometric structure across all layers. When they are imbalanced, the stream collapses to rank-1.

This regularity is robust across variations in activation function, model scale, and normalization type, suggesting it is a *notable* property of the transformer architecture. It opens new avenues for architecture design, pruning, and training stabilization based on spectral balance rather than heuristic tuning.

Acknowledgments

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A Detailed Methodology

A.1 Weight Extraction

For each model, we load the pretrained weights from HuggingFace and extract:

- **MLP weights:** For SwiGLU, W_{up} and W_{down} from the gate-up fused projection (splitting the fused matrix into gate and up components). For GELU, W_{up} and W_{down} from the standard FFN.
- **Attention weights:** The output projection W_o from each attention block. For GQA, we compute $W_{\text{attn}} = W_o W_o^\top$ to obtain a $[d, d]$ matrix for the covariance propagation.

A.2 Lyapunov Iteration

Starting with $C_0 = \mathbf{I}_d$, we iterate:

$$C_{\ell+1} = \frac{A_\ell C_\ell A_\ell^\top}{\|A_\ell C_\ell A_\ell^\top\|_2},$$

where $A_\ell = \mathbf{I} + W_{\text{mlp}}^{(\ell)} + W_{\text{attn}}^{(\ell)}$. The spectral norm normalization preserves the condition number and effective rank while preventing overflow.

A.3 Computational Details

All computations use double-precision floating point (float64) for the covariance matrices to handle the extreme condition numbers (10^{12}). Eigenvalue decompositions use `torch.linalg.eigvalsh` with adaptive thresholds for numerical zero detection.

A.4 Code Availability

The full analysis code, including the Lyapunov analyzer, architecture inspector, and plotting utilities, is available at <https://anonymous.github.io/>.

B Sensitivity Analysis

B.1 Normalization Strategy

We tested three normalization strategies for the Lyapunov iteration: trace normalization, spectral norm normalization, and Frobenius norm normalization. All three yield qualitatively identical results (same ranking of models by stability, same collapse layers), confirming that the 1:1 Rule is robust to the choice of normalization.

B.2 Attention Approximation

We compared two ways of incorporating attention:

1. $W_{\text{attn}} = W_o$ (direct output projection, valid for MHA)
2. $W_{\text{attn}} = W_o W_o^\top$ (covariance contribution, valid for GQA)

Both approximations yield the same qualitative conclusion: Qwen is stable, others collapse. The ratio values differ in magnitude but maintain the same ordering.

B.3 Excluding Attention

When attention is excluded from the Lyapunov iteration ($A = \mathbf{I} + W_{\text{mlp}}$), all models collapse immediately (layer 1–2). This confirms that attention plays a *critical stabilizing role* in the residual stream geometry.